Investments
Autumn 2026

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Problem Set #2 Mean-Variance Analysis

This problem set is due before class Week 3. You are to submit solutions to assignment using Canvas. You may work on this in groups of your choosing. Groups may be of no more than 5 people. Submit one solution for each group following the instructions on Canvas.

1. Obtain the data file **ps2_problem1.xlsx** from Canvas. In this file, the first column contains the date. Subsequent columns contain the returns to the *a)* Value Weighted ("Market") index of exchange listed stocks, *b)* Target *c)* Tesla and *d)* Alphabet common stocks. In all cases, the return is the simple return (one month) over the period from the previous date up to the date in column 1. (Note: the return for the first row (date 30JULY2010) is from 06/30/2010 to 07/30/2010).

   (a) Find the average returns on the Market and each of the three stocks over the entire period from July 30, 2010 through the end of 2025.

   (b) Find the variance of each of these returns over the entire period.

   (c) Calculate the covariance between the return on Target and the return on Tesla. Calculate the covariance between the return on Target and the return on Alphabet. Calculate the covariance between the return on Tesla and the return on Alphabet.

   (d) Consider the following two portfolios:

   - Portfolio A had 60% invested in Target, 20% in Tesla and 20% in Alphabet at the beginning of each month
   - Portfolio B had 60% invested in Tesla, 20% in Target and 20% in Alphabet at the beginning of each month.

i. Using just the average returns and variances of the three stocks, plus the covariances between the stocks (calculated in parts a, b and c), calculate the average returns and variances of these two portfolios, and the covariance between the two portfolios.

ii. Now on a spreadsheet calculate the realized returns of each the portfolios for each month. Using these realized returns calculate the portfolio average returns, variances and the covariance directly, confirming your answers from above.

2. The Triad family of mutual funds allows investors to split their money between several portfolios managed by Triad. Portfolio A consists entirely of risk-free securities, and has a certain return of 4%. Portfolio B has an expected return of 19% and a standard deviation of return of 25%. Portfolio C has an expected return of 10% and a standard deviation of return of 15%. Your client is leaning towards investing his money entirely in portfolio C, since he is unwilling to take the higher risk associated with portfolio B, but wants a higher return than offered by portfolio A.

   (a) In your role as a Triad investment advisor, you suggest to him an alternative portfolio (consisting of a combination of A and B) that has the same standard deviation as portfolio C but a higher expected return. If he has $200,000 to invest, how much should he invest in B and how much in A? What is his expected return in this case? (Note that since you work for Triad you cannot recommend investing in a market index fund, since this is not sold by Triad).

   (b) Sketch a risk/return diagram to help explain to him why his combination of risk and return is improved by this option, and why his risk has not increased. Show all four portfolios on the diagram (the original three plus your suggestion).

   (c) Now assume that the correlation between portfolios B and C is 0. Find the combination of B and C that is mean-variance efficient. That is find the "MVE" portfolio that combines these two portfolios. Calculate and report the portfolio weights in the MVE portfolio. Also calculate the expected return and standard deviation of the MVE portfolio. What is the Sharpe ratio of the

MVE portfolio? How does this compare to the Sharpe ratio for portfolios B and C?

(d) Using the "MVE" portfolio and portfolio A find a portfolio that has the same standard deviation as portfolio C but has higher expected return. What amount of money would be invested in each of the securities A, B and C to achieve this return? How does the expected return of this portfolio compare with the expected return of the portfolio you constructed in part 1a? Why is it different?

(e) Draw a diagram that shows how considering all possible combinations of investments in the available portfolios improves the set of risk and return choices. Indicate the set of combinations that investors should prefer (e.g., those which minimize risk for a given level of return).

3. Consider the following assumption about annual returns to two stock mutual funds "A" and "B."

$$\begin{array}{ll} E(r_A) = 8\% & \sigma_A = 20\% \\ E(r_B) = 12\% & \sigma_B = 30\% \\ \rho_{A,B} = 0.2 & \end{array}$$

In addition a one-year T-Bill offers a risk-free return of 2%.

(a) You have $1 million to invest. Although we know that diversification is important, let's consider investments in just one risky security along with the T-Bill. Suppose that you have a target expected rate of return of 10% for your portfolio.

i. How could you achieve this target by investing in the T-Bill and mutual fund A? What percentage and dollar investment would be in the T-Bill and in the mutual fund? What is the standard deviation of your portfolio return? What is the Sharpe ratio of your portfolio? How does this compare to the Sharpe ratio of mutual fund A?

ii. Instead how could you achieve the target by investing in the T-Bill and mutual fund B? What percentage and dollar investment would be in the T-Bill and in the mutual

fund? What is the standard deviation of your portfolio return? What is the Sharpe ratio of your investment? How does this compare to the Sharpe ratio of mutual fund B?

(b) Both of the portfolios above achieve an expected return of 10%. Which is preferred? Why?

(c) Suppose now that you can invest in a combination of the T-Bill, mutual fund A and mutual fund B. Once again your objective is achieve an expected return of 10%. How would you invest your money? Be specific (percentage investments and dollar investments please). What is the Sharpe ratio of your portfolio? How does it compare to the Sharpe ratios of mutual funds A and B?